THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The magnetic field produced by a straight wire at a distance 4 cm is B. At 2 cm from the wire (same current), the field is 1

(a) larger than B **(b)** zero **(c)** equal to B **(d)** smaller than B

2. Speed of light in a medium of n = 2 is 1

(a) 1.5×10^8 m/s **(b)** 3×10^8 m/s **(c)** 2×10^8 m/s **(d)** 6×10^8 m/s

3. The deviation of violet light in a prism is greater than that of red because 1

(a) red is absorbed
(b) the refractive index for violet is higher
(c) violet travels faster in glass
(d) violet has lower frequency

4. A 10 ohm and a 40 ohm resistor are in parallel. The equivalent resistance is 1

(a) 5 ohm **(b)** 25 ohm **(c)** 50 ohm **(d)** 8 ohm

5. An alpha particle moving north is deflected towards the east in a magnetic field. The force is 1

(a) zero **(b)** against the velocity
(c) along the velocity **(d)** perpendicular to the velocity

6. An object 3 cm high is at 20 cm from a convex lens of f = 10 cm. The image height is 1

(a) 3 cm (erect) **(b)** 6 cm **(c)** 1.5 cm **(d)** 3 cm (inverted)

7. The Sun at noon appears white because 1

(a) violet is absorbed
(b) light travels a short path and little blue scatters
(c) red gets scattered
(d) no scattering occurs

8. A 100 W bulb has a resistance at 220 V of 1

(a) 22 ohm **(b)** 2.2 ohm **(c)** 100 ohm **(d)** 484 ohm

9. The induced current in a coil is zero when 1

(a) the field changes
(b) the magnet is at rest relative to the coil
(c) the coil is moved
(d) the magnet is moved

10. An object is placed 18 cm from a concave mirror of f = 12 cm. The image distance is 1

(a) 18 cm **(b)** 24 cm **(c)** 36 cm **(d)** 12 cm

11. A lens of power -2.5 D has focal length 1

(a) -25 cm **(b)** +40 cm **(c)** -40 cm **(d)** -2.5 cm

12. The work done in moving 4 C of charge across a potential difference of 9 V is 1

(a) 13 J **(b)** 36 J **(c)** 2.25 J **(d)** 5 J

13. Looking at a circular coil, the current is anticlockwise. The face seen behaves as 1

(a) a south pole **(b)** neutral **(c)** a north pole **(d)** both poles

14. Light enters from air into a medium at an angle of 45 degrees and refracts at 30 degrees. The refractive index is 1

(a) $\sqrt{3}$ **(b)** $\sqrt{2}$ **(c)** 2 **(d)** $1/\sqrt{2}$

15. In glass, the colour of light with the highest speed is 1

(a) red **(b)** green **(c)** blue **(d)** violet

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a person with hypermetropia hold a book farther away to read? 2

17. A 20 ohm resistor is connected in parallel with a 5 ohm resistor. Find the equivalent resistance. 2

18. A current-carrying wire is parallel to a uniform magnetic field. State the force and give the reason. 2

19. Why can a concave lens never form a real image? 2

20. Why is the sky above a mountain-top sometimes darker blue? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. State three differences between an electric motor and an electric generator. 3

22. Differentiate between the images formed by a concave mirror and a convex lens for an object at 2F. 3

23. What is the minimum distance at which objects can be seen clearly? Why do we hold things farther than a few cm? 3

24. How many 220 ohm resistors must be connected in parallel to draw a total current of 5 A from a 220 V line? 3

OR

What is the difference between the magnetic field due to a straight wire and that due to a solenoid? 3

25. A convex mirror has a focal length of 20 cm. An object is placed 20 cm in front of it. Find the image distance and magnification. 3

OR

A hypermetropic person has a near point at 75 cm. Find the power of the lens to read at 25 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The refractive indices of kerosene, water, glass and diamond are 1.44, 1.33, 1.5 and 2.42. **5**

(a) Arrange in increasing order of optical density

(b) find the speed of light in diamond

(c) state in which light bends the most.

OR

A person's far point is at 80 cm and near point at 25 cm. Find the power of the lens needed to see distant objects. Is any lens needed for reading? **5**

27. Explain Joule's law of heating. Find the heat produced when a current of 2 A flows through a 50 ohm resistor for 10 minutes. State the heat in calories (1 cal = 4.2 J). **5**

OR

A household circuit has a 5 A circuit for lights and fans and a 15 A circuit for power appliances. Describe its layout, state why a geyser of 2 kW needs the 15 A circuit and why a fuse is used. **5**

END OF QUESTION PAPER — **Total: 50 Marks**

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